

Assignment - Data Transform Activity (20%)

Due: Module 4

Purpose:

The purpose of this assignment is to transform data into an appropriate format for analysis and then extract that data for further problem solving.

Assignment Objective:

For the business problem you selected, you will extract data for further analysis using a series of transformations, including: select, mutate, filter, and arrange.

Instructions:

This activity relates to the Video Games Sales Data:

Click on the link below to download your data:

- [Video Games Sales - Data](#)

In this assignment, you will perform simple transformations on your data (you might need to reload the data in HDInsight and Hive). Do the following steps:

- 1) a) Create a new table for your analysis call "sales_genre".
b) Load the table "sales" into this table.
c) Select these columns: Genre, Global_Sales, Critic_Score.
- 2) Round the data found in the "Global_Sales" column. (HINT: the SQL function to round a number is ROUND(obs)
- 3) Filter the data to only look at those items in the "Critic_Score" column that are greater than 0.
- 4) Order the data in the "Critic_Score" column from highest to lowest. (HINT: Use the DESC query)

This assignment relates to the following Course Learning Requirements:

CLR 3: Recommend which tools and the best use of these tools, to support data analysis, data quality, problem solving, analytics and business decision machining.

CLR 4: Extract and load data to support problem solving and decision-making.

Grading for This Assignment

Steps required	Grade Earned
1. a) Create a new table for your analysis call "sales_genre". b) Load the table "sales" into this table. c) Select these columns: Genre, Global_Sales, Critic_Score. 2. Round the data found in the "Global_Sales" column. (HINT: the SQL function to round a number is ROUND(obs)	50%
3. Filter the data to only look at those items in the "Critic_Score" column that are greater than 0.	80%
4. Order the data in the "Critic_Score" column from highest to lowest. (HINT: Use the DESC query)	100%

1A)

```
CREATE TABLE sales_genre
(
  Genre string,
  Global_Sales decimal(10,2),
  Critic_Score int
);
```

The screenshot shows the Hive web interface. At the top, there's a navigation bar with tabs for QUERY, JOBS, TABLES, SAVED QUERIES, UDFs, and SETTINGS. On the right, there are buttons for '+ NEW JOB' and '+ NEW TABLE', and a 'NOTIFICATIONS' bell icon. Below the navigation bar, there's a 'Worksheet1' tab. The main area is divided into two sections. The left section is labeled 'DATABASE' and contains a text input for 'Select or search database/schema' with a dropdown menu showing 'default'. The right section is labeled 'Tables(5)' and contains a list of tables: 'hivesampletable', 'sales', 'sales_genre', 'survey', and 'survey_consequences'. The 'sales_genre' table is highlighted. In the background, a SQL query is visible in the editor:

```
1 CREATE TABLE sales_genre
2 (
3   Genre string,
4   Global_Sales decimal(10,2),
5   Critic_Score int
6 );
7
```

1B) & 1C)

```
INSERT OVERWRITE TABLE sales_genre
SELECT Genre, Global_Sales, Critic_Score
FROM sales;
```

HIVE

+

NEW JOB

+

NEW TABLE

QUERY

JOBS

TABLES

SAVED QUERIES

UDFs

SETTINGS

NOTIFICATIONS

Worksheet1 * x Worksheet3 * x +

DATABASE

Select or search database/schema

default

Browse

1 INSERT OVERWRITE TABLE sales_genre

2 SELECT Genre, Global_Sales, Critic_Score

3 FROM sales;

4

5

default ✓ Tables(5)

Search Tables

hivesampletable

sales

sales_genre

survey

survey_consequences

RESULTS FROM:

SELECT * FROM sales_genre LIMIT 10;

Worksheet1 * x Worksheet3 * x +

DATABASE

Select or search database/schema

default

1 SELECT * FROM sales_genre LIMIT 10;

✓ Execute

Save As

Insert UDF

Visual Explain

RESULTS

LOG

VISUAL EXPLAIN

TEZ UI

Filter columns

x

sales_genre.genre	sales_genre.global_sales	sales_genre.critic_score
Sports	82.53	76
Platform	40.24	null
Racing	35.52	82
Sports	32.77	80
Role-Playing	31.37	null
Puzzle	30.26	null
Platform	29.80	89
Misc	28.92	58
Platform	28.32	87
Shooter	28.31	null

2)

```
INSERT OVERWRITE TABLE sales_genre
SELECT Genre, ROUND(Global_Sales), Critic_Score
FROM sales_genre;
```

RESULTS FROM:

```
SELECT * FROM sales_genre LIMIT 10;
```

sales_genre.genre	sales_genre.global_sales	sales_genre.critic_score
Sports	83.00	76
Platform	40.00	null
Racing	36.00	82
Sports	33.00	80
Role-Playing	31.00	null
Puzzle	30.00	null
Platform	30.00	89
Misc	29.00	58
Platform	28.00	87
Shooter	28.00	null

3)

```
INSERT OVERWRITE TABLE sales_genre
SELECT *
FROM sales_genre
WHERE Critic_Score > 0;
```

The screenshot shows the Hive Query Editor interface. At the top, there's a navigation bar with tabs for QUERY, JOBS, TABLES, SAVED QUERIES, UDFs, and SETTINGS. Below this is a 'Worksheet1' tab. The main area is labeled 'DATABASE' and contains a text input field with the SQL query: `1 INSERT OVERWRITE TABLE sales_genre`, `2 SELECT *`, `3 FROM sales_genre`, `4 WHERE Critic_Score > 0;`. At the bottom, there are buttons for 'Execute', 'Save As', 'Insert UDF', and 'Visual Explain'.

RESULTS FROM:

```
SELECT * FROM sales_genre LIMIT 10;
```

The screenshot shows the Hive Query Editor interface with the results of a query. The query entered is `1 SELECT * FROM sales_genre LIMIT 10;`. Below the query editor, there are buttons for 'Execute', 'Save As', 'Insert UDF', and 'Visual Explain'. The 'RESULTS' tab is selected, showing a table with 3 columns: `sales_genre.genre`, `sales_genre.global_sales`, and `sales_genre.critic_score`. The table contains 10 rows of data. On the right side, there is a 'Search Tables' panel showing a list of tables: `hivesampletable`, `sales`, `sales_genre`, `survey`, and `survey_consequences`.

sales_genre.genre	sales_genre.global_sales	sales_genre.critic_score
Sports	83.00	76
Racing	36.00	82
Sports	33.00	80
Platform	30.00	89
Misc	29.00	58
Platform	28.00	87
Racing	23.00	91
Sports	23.00	80
Misc	22.00	61
Sports	22.00	80

4)

```
INSERT OVERWRITE TABLE sales_genre
SELECT *
FROM sales_genre
ORDER BY Critic_Score DESC;
```

HIVE + NEW JOB

QUERY JOBS TABLES SAVED QUERIES UDFs SETTINGS NO

Worksheet1 * +

DATABASE default Browse

```
1 INSERT OVERWRITE TABLE sales_genre
2 SELECT *
3 FROM sales_genre
4 ORDER BY Critic_Score DESC;
```

Execute Save As Insert UDF Visual Explain

RESULTS LOG VISUAL EXPLAIN TEZ UI

Filter columns x

default v

Search Tables

- hivesampletable
- sales
- sales_genre
- survey
- survey_consequences

RESULTS FROM:

```
SELECT * FROM sales_genre LIMIT 10;
```

Worksheet1 * +

DATABASE default Browse

```
1 SELECT * FROM sales_genre LIMIT 10;
```

Execute Save As Insert UDF Visual Explain

RESULTS LOG VISUAL EXPLAIN TEZ UI

Filter columns x

sales_genre.genre	sales_genre.global_sales	sales_genre.critic_score
Action	11.00	98
Action	11.00	98
Sports	5.00	98
Fighting	0.00	98
Action	5.00	97
Sports	4.00	97
Action	21.00	97
Shooter	3.00	97
Platform	11.00	97
Platform	8.00	97

default v

Search Tables

- hivesampletable
- sales
- sales_genre
- survey
- survey_consequences